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Analysing the effects of solar insolation and temperature on PV cell characteristics

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ABSTRACT

Shortage of power, degradation of the environment, and high cost associated with fossil-fuels like coal, natural gas, oil, etc. compel us to find some economical, sustainable and eco-friendly energy sources. Presently, solar energy is gaining worldwide acceptance fora long and is gathering people's attention as it is not only clean, pollution-free but also available on earth without any cost. The usage of solar energy in an efficient way is a big challenge for all researchers. The most commonly used materials for the fabrication of solar cells are mono-crystalline silicon, thin-film technology, polycrystalline silicon and nano-materials. These materials have selected because they are all either in production or have the expectancy of being in production over the next few years. In the present scenario, silicon material based PV cells are dominating the power industry. But in the coming time, nanomaterials such as graphene, carbon nanotubes and many others will be dominating the systems due to their small size and high electrical conductivity. Studying the PV cell characteristics is an important way to know that how a solar cell works and responding to various parameters. This study can be done either through fabrication or mathematical modeling. Hence, in this paper, a single diode solar cell model is chosen and the effect of intensity of solar insolation and temperature is seen. Further, the simulations are being performed to understand the IV and PV characteristics. Finally, the effect of variations in temperature and solar irradiance on electrical parameters such as s/c current, Fill-Factor, o/c voltage, and conversion efficiency have been also discussed.

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1. Introduction

The power crisis is a worldwide concern for all. To get rid from it in a sustainable and eco-friendly manner is a big challenge. There are various renewable energy resources like biomass, wind, solar, hydro-power, geothermal, etc that are gaining importance worldwide to meet energy demand. Among these resources, solar energy has emerged as a convincing and independent renewable energy technology as others have been derived from the sun. The idea to utilize solar energy was well received in society through the solar panels. Although, the usage of solar energy and its range of accomplishmentsvary from a small calculator, watch to largesmart-grid technology. If, this enormous amount of energy is used in the right way, can help to fulfill the power demand of the world [1]. A solar cell acts as a two-terminal semiconductor diode and has the ability to convert solar radiations directly into electricity via the photovoltaic effect. Therefore, a conventional solar cell is also known as a photovoltaic(PV) cell. Solar cells are joined together to make solar-module and further, these modules are joined to make solar-panels [2]. Panels can be joined together in either parallel or series combinations, or both to form solar arrays so that large power can be produced. The most commonly used materials for the fabrication of solar cells are mono-crystalline silicon, amorphous silicon, poly-crystalline silicon, thin-film technology, and various nano-materials. The selection of these materials hasa great impact on the performance and high-power conversion efficiency of solar cells. Currently, a mono-crystalline silicon-based solar cell is dominating the solar market because of its high-power conversion efficiency, long life-span, non-toxic and stable nature [3].Various studies have shown that nano-materials also gained primacy in the solar cell fabrication area due to their unique physical, electrical and chemical properties. Moreover, nano-materials exhibit

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excellent applications in other fields like automobiles-parts, cell-phones, biosensors, airbags, laser-technology, traffic-signals, fiber networks, laptops, car-taillight, satellite dishes, etc. [4]. So, nano-materials are the future replacement of silicon material due to their small size, tuning of bandgap, and high electrical conductivity. Hence, in this paper, mathematical modeling of a silicon-based solar cell using MATLAB is done by considering the effect of ambient temperature and sunlight intensity to analyze its characteristics as well as its performance. Further, the effect of variations in temperature and solar irradiance on electrical parameters such as s/c current, Fill-Factor, o/c voltage, and high-power conversion efficiency have been also analyzed.

2. Need for modeling of solar cell

Advancements in the solar-industries and expanded numbers of installed solar panels in all over the world raised the demand for supervision and simulation tools for solar systems. Inthe last few decades, a lot of work has done to flourishmany simulation models and has continuously updated for researchers to know the performance of PV systems in a better way [4]. Various mathematical models have been developed and these models differ, depending on the types of software used by researchers such as Excel, PSPICE, MATLAB /Simulink, C-programming, etc. Some researchers have also done modeling and testing of PV modules/arrays based on a microprocessor. Mathematical modeling plays an important role as it reduces fabrication costs, saves time, and helps the researchers to predict the behavior of the PV system. A simulation is a useful tool for many purposes such as to study and analyze the behavior of power converters connected with the PV system. Estimates PV system efficiency and analyze various operational and environmental conditions suchas solar insolation, temperature, and faulty conditions [5]. The PV modules data-sheet given by manufactures, only provide information on electrical parameters like s/c current, o/c voltage, the rated value of voltage and current at the maximum power point, s/c current, and o/c voltage temperature coefficients at standard test conditions (STC) which are as PV_cell reference temperature = 25 °C, solar spectrum density distribution = 1.5 (AM) and solar_irradiance = 1000 w/m^2. Therefore, models are necessary to build, to predict the behavior as well as the performance of the solar system at any operating and weather conditions.

2.1. Equivalent circuit of solar cell

A solar cell acts as a two-terminal semiconductor diode. when a sufficient amount of sunlight or some other light source energy exceeding the band-gap energy of the solar cell material, falls upon it and generate electron-hole pairs called mobile charge carriers [6,7]. These charge carriers are separated by the diode-like structure of the cell, which produces a potential difference and thus generates electricity. The IV characteristic of a p-n junction diode can be described by the Shockley equation given below:

$$I_d = I_{rs}[\exp\left(\frac{QV_d}{ckT_c}\right) - 1] \tag{1}$$

Where I_d = Forward biascurrent generated by the diode in Ampere
I_{rs} = Nomial value of diode reverse saturation current in Ampere
V_d = Voltage across the diode in Volts
Q = Charge on Electron in Coulombs
T_c = standard temperature of cell in Kelvin
c = Diode ideality factor (Dimentionless)
k = Boltzmann's constant

The reciprocal of the term (Q/ckT_c) is called diode thermal voltage and depends on standard temperature (T_c). Therefore

$$V_{th} = \left(\frac{ckT_c}{Q}\right)$$

Where V_{th} = Diode thermal voltage

Therefore, we can rewrite equation (1) as

$$I_d = I_{rs}[\exp\left(\frac{V_d}{V_{th}}\right) - 1] \tag{2}$$

Semiconductor diode behaves as a non-linear device and the above exponential equation is used to describe its basic fundamental characteristics. PV cell operating principle can also be described from the PN junction. The main factors that affect the performance of the solar system are solar insolation, surrounding temperature, localized climate conditions, and raw materials properties[8,9]. There are many solar cell models described in the literature to analyze its actual behavior. Usually, a PV cell is represented by a current source (I_{ph}) called cell photocurrent, one, two, or three diodes in parallel with the current source, an intrinsic shunt resistance (r_{sh}) or series resistance(r_s) or both connected. In this paper, the equivalent circuit of a practical solar cell using a single diode is shown in Fig. 1 below:

When we apply KCL on the above equivalent circuit, we get :

$$I_{ph} = I_d + I_{sh} + I_L \tag{3}$$

Here, I_d = Forward bias current generated by the diode in Ampere
I_{Sh} = Current flowing through parallel resistance (r_{Sh}) in Ampere
I_{ph} = photon current generated in a solar cell in Ampere
I_L = Load or output current of the solar cell in Ampere
T_c = standard temperature of cell in Kelvin
G = solar radiation in W/m^2

Equation (3) representsthe solar cell operating principle and can write the load or output current equation as follows:

$$I_L = I_{ph} - I_d - I_{sh} \tag{4}$$

Put $I_d = I_{rs}\left[\exp\left(\frac{V_d}{V_{th}}\right) - 1\right]$ and $I_{sh} = \frac{(V_L + I_{L}r_s)}{r_{sh}}$ in equation (4) we get,

$$I_L = I_{ph} - I_{rs}\left[\exp\left(\frac{V_d}{V_{th}}\right) - 1\right] - \frac{(V_L + I_L r_s)}{r_{sh}}$$

Or

$$I_L = I_{ph} - I_{rs}\left[\exp\left(\frac{V_L + I_L r_s}{V_{th}}\right) - 1\right] - \frac{(V_L + I_L r_s)}{r_{sh}} \tag{5}$$

The above equation mathematically describes the IV characteristics of the single diode solar cell. The value of photon current (I_{ph}) generated in a solar cell is directly proportional to solar irradiance, surrounding temperature and given as [10,11]:

$$I_{ph} = [I_{SC} + K_{SCI}(\Delta T)] * \frac{G}{G_n} \tag{6}$$

Here, I_{ph} = Photon current generated in a solar cell in Ampere

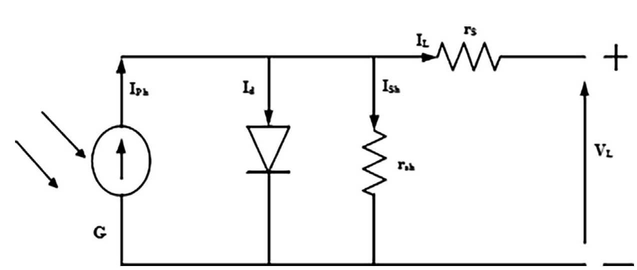

Fig. 1. Equivalent circuit ofa practical solar cell using a single diode.

I_{SC} = s/c current of the cell at STC in Ampere
K_{SCI} = s/c current temperature coefficient of cell
$\Delta T = (T_o\text{-}T_c)$
T_c = standard temperature of cell in Kelvin
T_o = Actual temperature of cell in Kelvin
G = Actual solar radiation in w/m^2
G_n = standard solar radiation in w/m^2

$$V_{oc} = [V_{ocs} + K_{ocv}(\Delta T)] \tag{7}$$

Here, V_{ocs} = Output or o/c voltage of cell at STC in Volts
K_{ocv} = o/c voltage temperature coefficient of cell
$\Delta T = (T_o\text{-}T_c)$
T_c = standard temperature of cell in Kelvin
T_o = Actual temperature of cell in Kelvin

When a semiconductor diode operates in reverse bias mode, then reverse saturation current rises with temperature and it may be expressed as:

$$I_{rst} = I_{rs}\left(\frac{T_o}{T_c}\right)^3 * exp[\frac{QE_g}{cK}\left(\frac{1}{T_o} - \frac{1}{T_c}\right)] \tag{8}$$

Here,
E_g = Energy band-gap of the semiconductor material in electron volts (eV)
I_{rs} = Reverse saturation current at STCand it is given by the equation below:

$$I_{rs} = \frac{I_{sc}}{[\exp\left(\frac{V_{oc}}{n_s * V_{th}}\right) - 1]}$$

V_{oc} = Output or o/c voltage per cell in Volt
I_{SC} = Output or s/c current per cell in Ampere

Thus, the standard equation of output or load current of PV module is given as:

$$I_L = n_p * I_{ph} - n_p * I_{rs}[\exp(\frac{V_L + (I_L * r_s)}{n_s * V_{th}} - 1] - \frac{(V_L + I_L * r_s)}{r_{sh}}$$

Efficiency in a photovoltaic cell is, measured by its ability to directly convert solar energy into DC energy. Therefore,the basic equation of efficiency (η)and Fill Factor (FF) of a solar cell is given below:

$$FF = \frac{(V_{mp} * I_{mp})}{(V_{oc} * I_{SC}\}} \text{and} \eta = \frac{P_{max}}{P_{in}}$$

2.2. Solar cell characteristics

Fig. 2 above shows the current–voltage(IV) and power-voltage (PV) curve of a particular silicon PV cell. IV curve represents a graph between the output current and output voltage under normal temperature and solar irradiance. The above characteristics curves give the necessary information required to compose maximum power conversion efficiency solar system that can operate as near to its maximum power point(MPP) as shown.When no-load is connected across the load terminals as shown in Fig. 1., the maximum amount of load voltage will generate across the output terminals and it is called o/c voltage (v_{OC}). At the other extreme, when load terminals are short-circuited means ideally load resistance value is zero, the maximum load current will flow through the load and this current is called s/ccurrent (I_{sc}). So, we can say that the maximum value of voltage called o/c voltage (v_{OC}) is obtained from open-circuit and the maximum value of current called s/ccurrent (I_{sc}) is obtained from short-circuit condition. The above two conditions generate no or zero electrical power, so there must be one particular value of voltage and current where the solar cell can generate maximum power. These points are I_{mp} and V_{mp}shown in Fig. 2 [3,4]. The shape of pink shaded area under the current–voltage characteristics curve should be perfect-square for an ideal solar cell, which shows the optimum use of a particular solar cell. But, practical solar cell didnot have a perfectly square shape, because solar cell output current and voltage both depend on changes inthe intensity of solar light and ambient temperature.

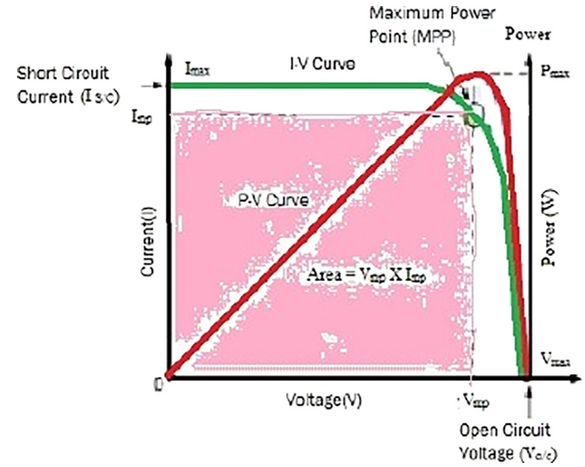

Fig. 2. IV and PV Characteristic of solar cell [3].

2.3. Factorsaffecting solar cell performance

PV cell is the main unit in the solar power generation system. There are various parameters thataffect the solar cell performance and these are:

- Solar insolation: The photon current 'I_{ph}' is directly proportional to solar irradiance as shown by equation (6). So, as solar radiation increases, the photon current also increases. On the other hand, the output or o/c voltage of a solar cell is less affected by it [8,9].Therefore, with an increase in solar radiation, photon current increases which in turn increases output power and hence its efficiency.
- Temperature: Temperature is also an important factor and has a great impact on the reverse saturation current as shown in equation(8). As temperature increases, reverse saturation current also increases, which leadsto a reduction in output or open-circuit voltage and maximum useful power [10,11].Various study showsthat with per degree increase in temperature output voltage decreases by 2.2 mV, useful power by 0.45 and efficiency by 0.06.
- Ideality factor: The ideality factor 'c' is a measure of the material quality and tells how closely the practical diode behaves like an ideal diode. The ideal value of the diode ideality factor is unity but its practical value lies between 1 and 2. The low value of 'c' means better material, a small value of reverse saturation current, and high output power.
- Shunt and series resistance: Series resistance is the ohmic resistance in solar cells and it comes due to metallic contact. On the other hand, we have shunt resistance and it is due to a crystallographic defect in the solar material. The value of series resistance should be zero and shunt resistance be infinity for an ideal solar cell. Practically, the low value of shunt resistance

Table 1
Electrical parameters of the DS-100 M solar power system.

Parameters	Values
Rated power	100 Watts
Rated voltage (V_{mp})	18 V
Rated current (I_{mp})	5.55A
o/c voltage (V_{oc})	21.6 V
s/c current (Isc)	6.11A
Total cells connected in series (n_s)	36
Total cellsconnected in parallel (n_p)	1
Maximum system voltage	1000 V
Range of operating temperature	−40 °C to 80 °
IscTemperature coefficient	0.002%/°C

and high value of series resistance leading to high power loss, low Fill-Factor, and low efficiency. Therefore, for high efficiency, all practical solar cells must havea small value of series resistance anda large value of shunt resistance.

3. Result and discussion

The solar cell can be simulated with an equivalent circuit model as shown in Fig. 1 [13] and for this, a solar system of 100 W is taken as the reference module. Its detailed electrical parameters data sheet is given below in Table1 [13,14].

The above module used the fundamental circuit equations of a solar cell explained above to describe the effects of environmental parameters such astemperature and intensity of solar light. MATLAB is used to plot the characteristics curves with different solar radiations at constant temperature and with variable cell temperature at fixed solar irradiance to analyze its performance.

3.1. Effect of changing the solar radiations keeping the constant temperature on cell performance:

Solar radiations havea great influence on the photon current (Iph) of PV cell. When solar light intensity increases from 200 w/m^2 to 1000 w/m^2, the load or s/ccurrent (I_L) increases as shown in Table 2 above. The IV and PV characteristics curve of the solar cells is shown below in Figs. 3 and 4 respectively. Therefore, with an increase in solar radiations, photon current increases which in turn increases the output power of the solar cell. On the other hand, the intensity of solar light has little effect on the output or o/cvoltage(V_L) [8,13,14].

3.2. Effect of variation in temperature keeping constant solar insolation on cell performance:

The IV and PV characteristics curve of a solar cell with changes in temperature keeping fixed solar insolation is shown below in Figs. 5 and 6 respectively. The IVcurve shows that, as temperature increases, o/c voltage reduces and its corresponding maximum output power also decreases. The PV curve showsthe similar change as in solar cell power when the light intensity remains constant [12–14].

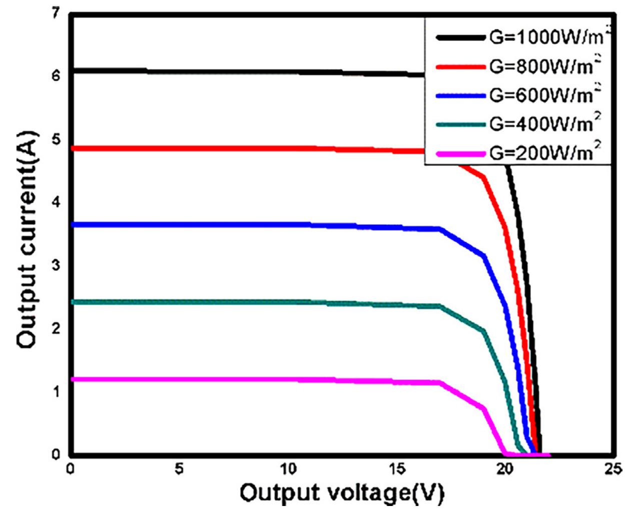

Fig. 3. IV Characteristic curve of solar cell.

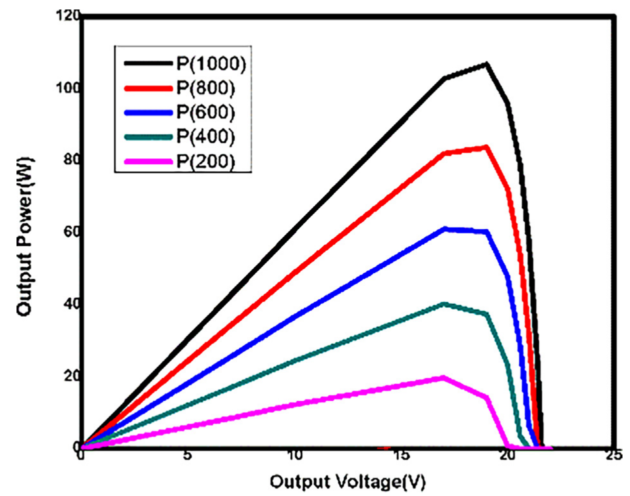

Fig. 4. PV Characteristic curve of solar cell.

Table 2
Load current (I_L) for different values of output voltage with variable solar irradiance at constant temperature.

S. no	I_L (Amp) (1000 W/m^2)	I_L (Amp) (800 W/m^2)	I_L (Amp) (600 W/m^2)	I_L (Amp) (400 W/m^2)	I_L (Amp) (200 W/m^2)
1	6.11	4.888	3.666	2.444	1.222
2	6.11	4.887	3.66	2.44	1.22
3	6.10	4.887	3.66	2.44	1.22
4	6.10	4.88	3.66	2.44	1.22
5	6.10	4.88	3.66	2.44	1.22
6	6.10	4.87	3.65	2.44	1.21
7	6.04	4.82	3.59	2.37	1.155
8	5.63	4.41	3.19	1.970	0.74891
9	4.8	3.6224	2.400	1.17	0.043
10	2.75	1.5051	0.28	0.00	—————
11	0.00	0.00	——————	——————	——————

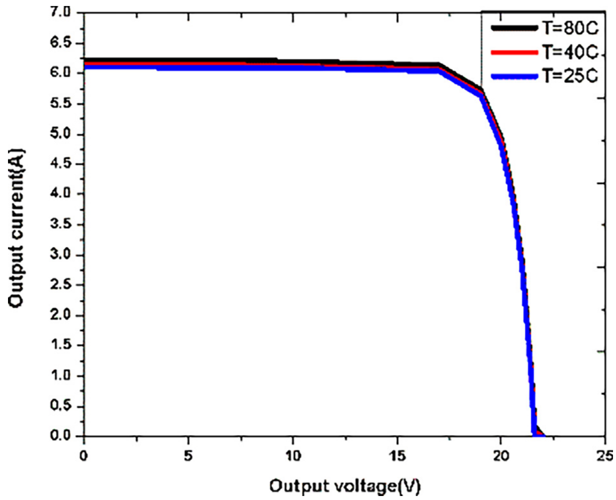

Fig. 5. V-I Characteristic curve of solar cell.

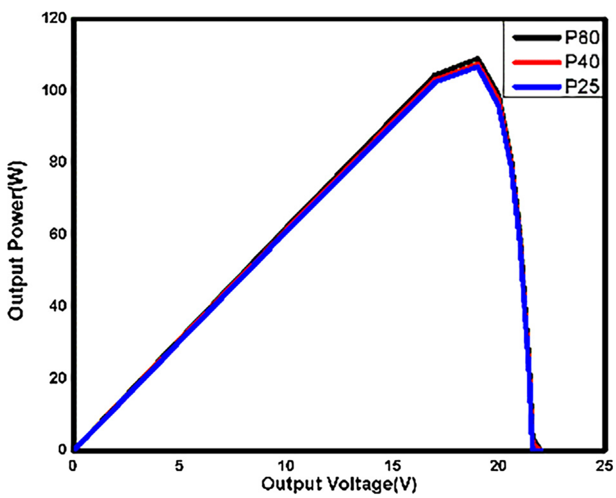

Fig. 6. P-V Characteristic curve of solar cell.

4. Conclusion

A solar cell is the main unit in the solar power generation system. It is a clean, green, and independent source of renewable energy. Therefore, the demand for solar energy is growing faster than ever before. To analyze the performance of the reference model under different temperature andsolar insolation, a specific solar model is used. The above graphs show that when the solar irradiance increases at a constant temperature, the output current increases, which leads to an increase in output power. Similarly, astemperature changes for constant irradiance, the output voltage decreases. Which results in a decreasein output power and therefore inefficiency. Finally, simulation results showa strong nonlinearity behavior of the PV cell.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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